

Topological Spectral Susceptibility in Three-Dimensional Elastic Networks

Edge-Level Spectral Leverage, Isospectral Controls, and Synthetic-Parameter Subspace
Transport

Topological-Quantum-Shielding Project

Version 9.2.1 – Final Mainline

Abstract

This work develops a computational framework for studying how three-dimensional network structure, elastic-network eigenmodes, edge-level spectral sensitivity, controlled perturbations, and synthetic cyclic parameterization shape the response of a network model. The central quantity is an edge spectral leverage constructed from first-order eigenvalue sensitivity with respect to individual edge weights. The framework combines a degree-preserving Maslov–Sneppen null model, targeted versus matched-random edge ablation, an isospectral counterfactual that preserves eigenvalues while modifying eigenvector organization, and a synthetic parameter family $H(\theta)$ used to analyze modal-subspace transport and cycle persistence. The current v9.2.1 implementation is intentionally conservative in interpretation: it establishes a mathematical and numerical test framework, not a demonstration of a physical four-dimensional topological phase or a microscopic biological “quantum-shielding” mechanism. The principal hypothesis investigated is that the global eigenvalue spectrum may be an incomplete descriptor of dynamical structure when eigenvector geometry and edge-resolved spectral susceptibility are relevant.

1 Introduction

Elastic Network Models (ENMs) provide a reduced representation of biomolecular structure in which nodes are connected by effective elastic interactions. For a symmetric Hessian H , the normal modes satisfy

$$Hv_k = \lambda_k v_k, \quad (1)$$

where λ_k denotes an eigenvalue and v_k the corresponding eigenvector.

A standard spectral description focuses on the set of eigenvalues $\{\lambda_k\}$. The present framework asks a stronger structural question:

Can two systems with similar or identical eigenvalue spectra nevertheless possess measurably different response because their eigenvectors, modal subspaces, and edge-level spectral sensitivities differ?

This motivates a shift from a spectrum-only description toward a *spectral susceptibility* description in which the response of individual modes to individual network edges becomes an explicit observable.

The current project is organized as a sequence of increasingly structured computational tests:

3D ENM $\rightarrow (\lambda_k, v_k) \rightarrow$ Edge Spectral Leverage $\rightarrow$ Topology Null $\rightarrow$ Isospectral Test $\rightarrow H(\theta) \rightarrow$ Subspace Trajectory

(2)

2 Research Objective and Hypothesis

The principal hypothesis is

$$\boxed{\text{The global eigenvalue spectrum may be an incomplete descriptor of dynamical structure.}} \quad (3)$$

Operationally, the hypothesis is tested by separating three classes of information:

1. eigenvalue information, $\{\lambda_k\}$;
2. eigenvector or modal-subspace organization, $\{v_k\}$ and P_K ;
3. edge-resolved spectral sensitivity, $\{L_e\}$.

A positive result would not by itself establish a new physical phase. It would instead show that spectral eigenvalues alone do not exhaust the response information contained in the modeled operator.

3 Three-Dimensional Elastic Network Model

Let the node coordinates be

$$r_i \in \mathbb{R}^3, \quad i = 1, \dots, N. \quad (4)$$

For an edge $e = (i, j)$, define the normalized contact direction

$$u_{ij} = \frac{r_j - r_i}{\|r_j - r_i\|}, \quad (5)$$

and its directional projector

$$P_{ij} = u_{ij} u_{ij}^T. \quad (6)$$

The current v9.2.1 implementation uses a synthetic three-dimensional folded scaffold with $N = 100$, a 7 Å contact cutoff, and a unit spring constant. The Hessian is constructed as a $3N \times 3N$ symmetric matrix. The six rigid-body modes are removed before selecting the positive elastic modes used for analysis.

The resulting eigendecomposition is

$$H = V \Lambda V^T, \quad (7)$$

with orthonormal eigenvectors for the symmetric operator.

3.1 Numerical consistency conditions

The implementation explicitly checks that the Hessian is positive semidefinite within numerical tolerance and that six rigid-body modes are identified before constructing the positive-mode spectrum. These checks are numerical safeguards and should not be interpreted as experimental validation of the ENM itself.

4 First-Order Spectral Sensitivity

For an edge weight w_{ij} , first-order perturbation theory for a non-degenerate eigenvalue gives

$$\frac{\partial \lambda_k}{\partial w_{ij}} = v_k^T \frac{\partial H}{\partial w_{ij}} v_k. \quad (8)$$

For the directional ENM construction, this becomes

$$\boxed{\frac{\partial \lambda_k}{\partial w_{ij}} = (v_{k,i} - v_{k,j})^T P_{ij} (v_{k,i} - v_{k,j})} \quad (9)$$

and equivalently

$$\boxed{\frac{\partial \lambda_k}{\partial w_{ij}} = \left[u_{ij}^T (v_{k,i} - v_{k,j}) \right]^2} \quad (10)$$

This quantity links a particular network edge to the stiffness response of a particular eigenmode.

5 Edge Spectral Leverage

Let K denote the selected set of soft positive modes. The implementation currently uses the lowest ten positive modes. We define normalized inverse-eigenvalue weights

$$\alpha_k = \frac{\lambda_k^{-1}}{\sum_{q \in K} \lambda_q^{-1}}. \quad (11)$$

The aggregate edge spectral leverage is then

$$\boxed{L_e = \sum_{k \in K} \alpha_k \frac{\partial \lambda_k}{\partial w_e}} \quad (12)$$

Thus L_e provides an edge-level map of spectral susceptibility over the selected soft-mode sector.

Two concentration measures are used to summarize the leverage distribution. The Gini coefficient is computed from the non-negative leverage vector, while the edge participation ratio is represented here by

$$\text{IPR}_L = \sum_e p_e^2, \quad p_e = \frac{L_e}{\sum_f L_f}. \quad (13)$$

A high IPR_L corresponds to a more concentrated leverage distribution, while a lower value indicates broader participation.

6 Degree-Preserving Topology Null

To separate structure associated with the native graph from structure that can arise under a fixed degree sequence, the framework uses Maslov–Sneppen edge rewiring.

For every node i , the null model preserves

$$k_i^{\text{native}} = k_i^{\text{null}}. \quad (14)$$

The rewiring procedure rejects self-loops and duplicate edges. The implementation asserts equality of the complete degree sequence after randomization.

This is a degree-controlled topology null rather than a fully geometry-matched protein conformation. Because ENM edge projectors depend on the spatial orientation of each connected node pair, rewiring can alter the associated geometric projectors as well. Accordingly, the null model should not be interpreted as an isolation of topology from all geometric consequences.

7 Controlled Edge Ablation

The leverage field is converted into an intervention by selecting the top fraction of edges according to their mean leverage over the synthetic cycle. The default intervention fraction is five percent.

For a native graph with Hessian H , the targeted perturbation produces an ablated operator H_{top} . A matched-random ablation H_{rand} removes the same number of edges but chooses them randomly.

The primary spectral response is the shift in the lowest positive mode,

$$\Delta\lambda_{\min}^+ = \lambda_{\min}^{+, \text{ablated}} - \lambda_{\min}^{+, \text{native}}. \quad (15)$$

An ensemble of random ablations provides a null distribution against which the targeted intervention can be compared. In the current implementation, thirty random ablation runs are used.

The strongest version of the hypothesis predicts a systematic separation between high-leverage intervention and matched-random intervention. Such a separation must be assessed from repeated null samples rather than from one randomly chosen control realization.

8 Synthetic Parameter Dimension

Version 9 introduces a periodic synthetic control coordinate

$$\theta \in [0, 2\pi). \quad (16)$$

Rather than analyzing a single operator, the framework studies a family

$$H(\theta). \quad (17)$$

For an edge with phase ϕ_e , the current construction uses a periodic weight modulation of the form

$$w_e(\theta) = \max[1 + A \cos(\theta + \phi_e), \epsilon], \quad (18)$$

where A is the modulation amplitude and ϵ prevents non-positive edge weights. The default implementation uses $A = 0.12$ and samples twenty values of θ around the cycle.

At each value of θ , the model yields

$$\lambda_k(\theta), \quad v_k(\theta), \quad L_e(\theta). \quad (19)$$

The synthetic dimension is a mathematical parameterization of the operator family. It is not, by itself, evidence for a literal fourth spatial dimension or a physical four-dimensional topological material.

9 Modal-Subspace Geometry

Individual eigenvectors may undergo arbitrary sign changes and can mix within nearly degenerate eigenspaces. For that reason, the framework compares the subspace spanned by the first K positive modes through the projector

$$P_K(\theta) = V_K(\theta)V_K(\theta)^T. \quad (20)$$

A local subspace-transport metric is

$$D_{\text{sub}}(\theta) = \|P_K(\theta + \Delta\theta) - P_K(\theta)\|_F. \quad (21)$$

Principal angles between consecutive subspaces provide an additional basis-invariant diagnostic. These quantities measure how the selected soft-mode subspace moves through parameter space without relying on arbitrary signs of individual eigenvectors.

10 Isospectral Counterfactual

A key control experiment separates eigenvalue information from eigenvector organization. Suppose

$$H = U\Lambda U^T, \quad (22)$$

and let Q be orthogonal. Define

$$W = UQ, \quad (23)$$

so that

$$H_{\text{iso}} = W\Lambda W^T. \quad (24)$$

Because $Q^T Q = I$,

$$\text{spec}(H_{\text{iso}}) = \text{spec}(H), \quad (25)$$

while, in general,

$$W \neq U. \quad (26)$$

The current implementation rotates one low mode with a higher positive mode using a θ -dependent orthogonal transformation. Numerical checks compare the resulting eigenvalues with the native positive spectrum.

This construction is explicitly a *counterfactual symmetric-matrix control*; it is not required to correspond to a realizable protein contact graph. Consequently, the spatial edge quantity computed from its rotated modes is treated as a mode-coupling probe rather than as a literal ENM derivative $\partial\lambda/\partial w_e$ of the counterfactual graph.

11 Cycle Closure and Resolution Checks

A periodic parameterization should close numerically. The framework therefore evaluates

$$H(0) \approx H(2\pi). \quad (27)$$

The absolute closure error is defined as

$$\epsilon_{\text{abs}} = \max_{a,b} |H_{ab}(0) - H_{ab}(2\pi)|, \quad (28)$$

and a relative version is obtained by normalizing with the Frobenius norm of $H(0)$.

The parameter grid is also refined to test resolution stability. Previously observed smoke-test values for a smaller scaffold gave a relative cycle-closure error of approximately 2.2×10^{-17} and an isospectral eigenvalue error of approximately 1.7×10^{-14} . These are numerical consistency checks on the construction, not evidence of topological protection.

12 Cycle Persistence

At each θ , the top five percent of edges by leverage are marked. For an edge e , its cycle persistence is

$$\Pi_e = \frac{1}{N_\theta} \sum_{t=1}^{N_\theta} \mathbf{1}[L_e(\theta_t) \geq q_{0.95}(\theta_t)], \quad (29)$$

where $q_{0.95}(\theta_t)$ is the within-cycle 95th percentile of edge leverage.

A high Π_e indicates repeated membership in the highest-leverage subset. The implementation additionally tracks the standard deviation of leverage rank over the cycle as a measure of rank volatility.

13 Downstream Lindblad Response Model

The project retains a reduced four-dimensional spin Hilbert space for a $\text{FAD}^{\bullet-}-\text{Trp}^{\bullet+}$ radical-pair model. The basis is

$$\{|S\rangle, |T_+\rangle, |T_0\rangle, |T_-\rangle\}. \quad (30)$$

The density matrix evolves according to a Lindblad equation,

$$\dot{\rho} = -i[H_s, \rho] + \sum_m \left(L_m \rho L_m^\dagger - \frac{1}{2} \{L_m^\dagger L_m, \rho\} \right). \quad (31)$$

The current implementation uses three singlet-to-triplet jump channels and integrates the dynamics over a total time of 5 μs . Hermiticity, trace preservation, and positive-semidefinite density-matrix checks are enforced numerically during integration.

The coupling from network structure to the quantum layer is deliberately phenomenological. A dimensionless spectral score s is mapped monotonically to an effective rate according to a saturating closure of the form

$$\gamma_{\text{eff}} = \gamma_{\text{floor}} + \gamma_{\text{span}} \frac{s}{1 + s}. \quad (32)$$

Therefore,

$$\boxed{L_e \rightarrow \gamma_{\text{eff}} \rightarrow \rho(t) \rightarrow Y_S(t)} \quad (33)$$

is a model-level response bridge, not a microscopic derivation from biomolecular structure to a calibrated radical-pair transition rate.

14 Results and Current Scientific Status

The current repository establishes a continuous computational pipeline linking three-dimensional structure, modal analysis, edge sensitivity, intervention, isospectral control, and synthetic-cycle transport.

The strongest rigorously specified components are:

1. symmetric ENM Hessian construction and eigendecomposition;
2. first-order eigenvalue sensitivity with respect to individual edge weights;
3. edge spectral leverage over a selected soft-mode sector;
4. degree-preserving topology randomization;
5. targeted versus matched-random edge ablation;
6. basis-invariant modal-subspace comparison;
7. isospectral counterfactual construction;
8. periodic synthetic-parameter analysis and numerical cycle closure.

The central scientific interpretation is therefore methodological rather than sensational: the framework tests whether eigenvector geometry and edge-resolved susceptibility contain response information not represented by the eigenvalue spectrum alone.

15 Limitations

15.1 Synthetic structural scaffold

The current v9.2.1 benchmark is generated from a synthetic three-dimensional fold rather than directly from a measured biomolecular ensemble. Results at this stage should therefore be understood as method development and proof-of-concept rather than a biological claim.

15.2 ENM frequency interpretation

The raw Hessian eigenvalues in the present formulation are model stiffness quantities. They are not automatically experimentally calibrated absolute vibrational frequencies.

15.3 Topology null limitations

Degree-preserving rewiring controls the degree sequence but does not maintain the complete physical geometry of the native contact network.

15.4 Isospectral construction

The isospectral control is defined in matrix space and need not correspond to a realizable ENM contact graph.

15.5 Synthetic topology claims

Cycle closure, gap persistence, or nontrivial subspace transport alone do not establish a topological invariant, topological protection, or a nonzero second Chern number.

15.6 Quantum interpretation

The Lindblad layer is downstream and phenomenological. A stronger biological interpretation would require a microscopic vibronic or open-system derivation and independent calibration.

16 Falsification Logic

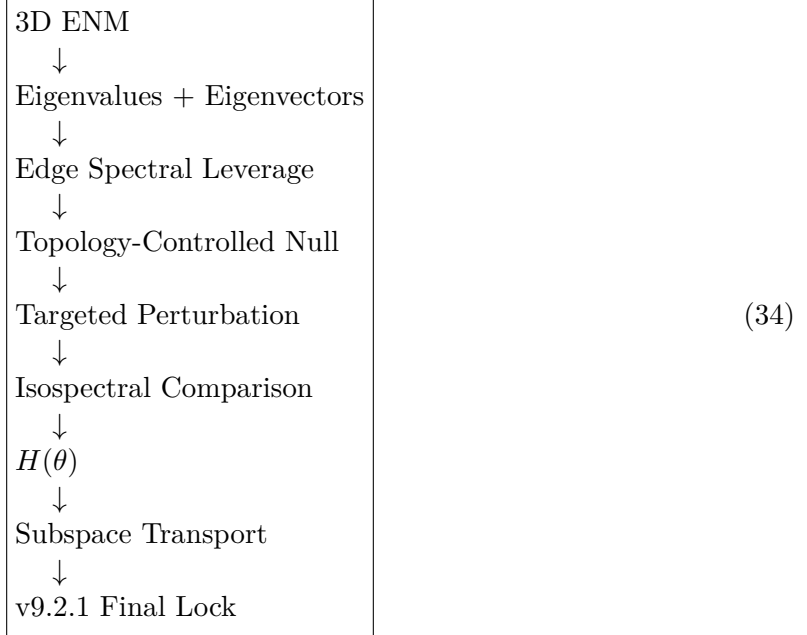
The framework is designed to admit negative outcomes. In particular, the central hypothesis would be weakened if, under appropriately matched controls,

1. targeted high-leverage ablations are not distinguishable from random matched ablations;
2. isospectral systems with altered eigenvector organization show no measurable response difference under the chosen observables;
3. leverage persistence disappears under increased parameter resolution or controlled perturbations;
4. apparent gap or transport effects are fully explained by numerical or normalization artifacts.

A successful result should therefore be reported together with the corresponding null distributions, effect sizes, numerical tolerances, and sensitivity to modeling choices.

17 Reproducibility and Repository Mainline

The current mainline is intentionally compact and is organized as five experiment scripts corresponding to the v8 through v9.2.1 development sequence. The intended conceptual order is



For reproducibility, the repository should retain the exact code version used for each reported result, the random seed, numerical tolerances, parameter values, and the corresponding output tables/figures when a final empirical dataset is available.

18 Conclusion

This work develops a computational framework for studying spectral susceptibility in three-dimensional elastic networks. The central mathematical object is the edge-resolved first-order spectral sensitivity

$$\frac{\partial \lambda_k}{\partial w_{ij}} = (v_{k,i} - v_{k,j})^T P_{ij} (v_{k,i} - v_{k,j}), \quad (35)$$

which is aggregated into an edge spectral leverage field over the soft-mode sector.

The resulting framework moves beyond a spectrum-only description by explicitly tracking eigenvectors, modal subspaces, edge leverage, controlled perturbations, and synthetic parameter transport. The isospectral construction provides a direct mathematical control for the proposition that identical eigenvalues do not necessarily imply identical structural organization of eigenvectors.

The most defensible current conclusion is therefore:

The framework provides a falsifiable computational route for testing whether spectral eigenvalues are an incomplete descriptor of response when eigenvector geometry and edge-resolved spectral susceptibility are taken into account.

The synthetic-dimensional and quantum-response extensions remain hypothesis-driven. They should not be interpreted as proof of a physical four-dimensional topological phase, a protected edge state, a second-Chern invariant, or a biological quantum-shielding mechanism.

Scientific Status Statement

Established within the model: ENM Hessian construction, symmetric eigendecomposition, first-order spectral sensitivity, edge spectral leverage, degree-preserving graph nulls, controlled

ablation, isospectral matrix construction, subspace-projector analysis, and periodic numerical closure checks.

Hypothesis-driven: persistent spectral textures, nontrivial spectral holonomy, topological protection, biological significance, and any microscopic coupling between network structure and quantum observables.

Code and Reproducibility

The computational source for the v9.2.1 analysis is maintained in the project repository:

<https://github.com/nujaneko-maker/Topological-Quantum-Shielding>